SSC CGL Advanced Practice Workbook – Set 2

Quantitative Aptitude (Advanced)

Algebra

1. If $x + 1/x = 5$, find $x^2 + 1/x^2$.
2. If $x - 1/x = 3$, find $x^3 - 1/x^3$.
3. Solve: $x^2 - 7x + 12 = 0$.
4. Solve: $2x^2 - 5x - 3 = 0$.
5. If $x+y=10$ and $xy=21$, find x^2+y^2.
6. Solve: $(x-2)(x-5)=14$.
7. Find roots of $x^2+8x+15=0$.
8. If $x^2+y^2=50$ and $xy=24$, find $x+y$.
9. Solve: $x^2-9=0$.
10. Find k if $x^2+kx+16$ has equal roots.

Geometry

1. Angles of a triangle are in ratio 2:3:4. Find all angles.
2. Area of a triangle = 216 cm^2, base = 18 cm. Find height.
3. Radius of circle = 14 cm. Find circumference.
4. Radius = 7 cm. Find area.
5. A chord subtends 90°. Find central angle.
6. Find diagonal of square with side 10 cm.
7. Perimeter of rectangle = 60 cm, length = 20 cm. Find breadth.
8. Area of rhombus with diagonals 12 cm and 16 cm.
9. Find area of equilateral triangle with side 12 cm.
10. Interior angle of regular octagon.

Mensuration

1. Volume of cylinder with r=7 cm, h=20 cm.
2. Curved surface area of cylinder.
3. Volume of cone with r=6 cm, h=14 cm.
4. Surface area of cube with side 8 cm.
5. Volume of sphere with radius 7 cm.
6. Difference between CSA and TSA of cylinder.
7. Water tank dimensions 5×4×3 m. Capacity?
8. Cube volume 1728 cm^3. Find side.
9. Radius doubled. By what factor does sphere volume increase?
10. Find TSA of hemisphere radius 14 cm.

Trigonometry

1. Find sin 30°.
2. Find cos 60°.
3. Find tan 45°.
4. If sin θ = 3/5, find cos θ.
5. Find sec 60°.
6. Evaluate $\sin^2\theta + \cos^2\theta$.
7. If tan θ = 4/3, find sec θ.
8. Find cot 45°.
9. Evaluate $(1+\tan^2\theta)$.
10. Height of tower if shadow = 20m and angle = 45°.

Data Interpretation

1. Company sales: 120,150,180,210,240. Find average.
2. Percentage growth from 200 to 260.
3. Pie chart sector = 90°. What percentage?
4. Total population = 50000. Sector 20%. Find population.
5. Revenue ratio 3:4:5. Total ₹120 lakh. Find each.
6. Average of 12,15,18,21,24.
7. Find median of 5,7,9,12,14.
8. Find mode of 2,4,4,5,6.
9. A bar graph shows production 400 and 500. Growth %?
10. Success rate 72%. Failures out of 1000?

Reasoning (Advanced)

Syllogism

1. All A are B. All B are C. Conclusion?
2. Some A are B. Some B are C. Conclusions?
3. No A is B. All B are C.
4. Some students are athletes.
5. All doctors are educated.
6. No teacher is lazy.
7. Some cars are electric.
8. All fruits are healthy.
9. Some books are novels.
10. All engineers are graduates.

Seating Arrangement

1. A,B,C,D,E sit in a row. A left of B. B left of C. Who is middle?
2. Six people sit around circle. Find opposite positions.
3. Eight friends face centre. Determine neighbour.
4. Linear arrangement with constraints.
5. Circular arrangement puzzle.
6. North-facing row puzzle.
7. South-facing row puzzle.
8. Mixed arrangement.
9. Double row arrangement.
10. Variable seating puzzle.

Coding-Decoding (Advanced)

1. FLOWER = GMPXFS, TREE = ?
2. COMPUTER = DPNQVUFS, SYSTEM = ?
3. If APPLE = 16642, MANGO = ?
4. Reverse coding pattern.
5. Position-value coding.
6. Symbol coding.
7. Matrix coding.
8. Alpha-numeric coding.
9. Word arrangement coding.
10. Multiple-step coding.

Logical Reasoning

1. Statement–Assumption.
2. Statement–Conclusion.
3. Cause–Effect.
4. Course of Action.
5. Assertion–Reason.
6. Strong/Weak Arguments.
7. Critical Reasoning.
8. Decision Making.
9. Data Sufficiency.
10. Analytical Puzzle.

English (Advanced)

Reading Comprehension

Read and answer:

"Technology has transformed communication. While distance once limited interaction, digital platforms now enable instant connectivity. However, concerns regarding privacy, misinformation and excessive dependence on technology continue to grow."

1. Main theme?
2. What changed communication?
3. One benefit mentioned?
4. One concern mentioned?
5. Synonym of transformed?
6. Antonym of instant?
7. Tone of passage?
8. Suitable title?
9. What is misinformation?
10. Summarize in one sentence.

Error Detection (Advanced)

1. Neither of the boys were present.
2. Hardly had he reached than it started raining.
3. One of my friend has cleared SSC.
4. No sooner did he arrive when the train left.
5. The furniture are expensive.
6. Mathematics are my favourite subject.
7. Each of the students have submitted.
8. He is senior than me.
9. She prefers tea than coffee.
10. I look forward to meet you.

Sentence Improvement

1. He do his work sincerely.
2. She have completed the task.
3. I am living here since 2015.
4. The news are shocking.
5. He is one of the best player.
6. She can sings well.
7. We discussed about the issue.
8. He denied to help.

9. I have seen him yesterday.
10. Scarcely had he entered when the bell rang.

Vocabulary

Synonyms

1. Pragmatic
2. Diligent
3. Benevolent
4. Meticulous
5. Ambiguous
6. Robust
7. Candid
8. Frugal
9. Elated
10. Resilient

Antonyms

1. Explicit
2. Optimistic
3. Scarcity
4. Expand
5. Fragile
6. Deteriorate
7. Hostile
8. Generous
9. Abundant
10. Triumph

SSC CGL Killer Topics

These topics repeatedly appear in higher-difficulty shifts:

1. Algebra Identities
2. Quadratic Equations
3. Geometry Theorems
4. Mensuration Formulas
5. Trigonometry Identities
6. Data Interpretation
7. Syllogism
8. Seating Arrangement
9. Reading Comprehension

10. Error Detection
11. Vocabulary
12. Current Affairs + Polity